

AYESHA ASLAM

PhD Candidate in *Traffic Information Engineering & Control*.

(Expected to graduate in January 2027)

Machine Learning & Deep Learning, Intelligent Transportation, Time-Series Forecasting, Android & Web Development, Teaching.

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Links: Portfolio | ORCID | Google Scholar | ResearchGate | LinkedIn

SUMMARY

I am a Ph.D. candidate in Traffic Information Engineering and Control at Chang'an University in Xi'an, expecting to graduate in January 2027. Before that, I completed an M.S. in Computer Science at Bahria University, Islamabad, and a B.S. in Computer Science at the University of the Punjab, Lahore, which gave me a strong foundation in computer science. My research focuses on four themes: autonomous vehicles, trajectory prediction, path planning, and unmanned systems, and my results have been published as journal articles, conference papers, and book chapters with Springer Nature, Elsevier, IEEE, and CRC Press. I have also taught computer science at schools in Islamabad for more than three years, supported by a Bachelor of Education. Outside academia, I like to sketch and paint in oils.

EDUCATION

- **Ph.D., *Traffic Information Engineering and Control*** **2022/09 – To be graduated in 2027/01**
  Chang'an University, Xi'an, China
- **Master of Science in *Computer Science (MSCS)*** **2019/09 – 2022/01**
  Bahria University, Islamabad, Pakistan
- **Bachelor of Education (B.Ed)** **2017/09 – 2019/01**
  Allama Iqbal Open University, Islamabad, Pakistan
- **Bachelor of Science in *Computer Science (BSCS)*** **2011/09 – 2016/01**
  University of the Punjab, Lahore, Pakistan

PROFESSIONAL EXPERIENCE

- **Computer Science Teacher** **2020/10 – 2022/11**
  Army Public School, Pothwar Campus, Islamabad, Pakistan
- **Teacher** **2019/03 – 2020/03**
  Dar-e-Arqam School, G-9/2, Islamabad, Pakistan
- **Teacher** **2018/09 – 2019/02**
  SLS School, G-11/4 Campus, Islamabad, Pakistan
- **HR Manager and Development Intern** **2017/09 – 2018/02**
  Itroos, Islamabad, Pakistan
- **Android Developer and Intern** **2017/03 – 2017/08**
  Xyper Technologies, Islamabad, Pakistan

SELECTED PUBLICATIONS

Refer to APPENDIX or Google Scholar for complete list of publications.

* Corresponding author; ** co-corresponding author.

1. **Ayesha Aslam**, Xiaojun Yang. Spatial-aware self-attention feature fusion DQN for path planning of USV in static and dynamic environments. Applied Soft Computing (2026), 115818, Elsevier.
2. **Ayesha Aslam**, Yang Xiaojun*, Kashif Naseer Qureshi**, Adil Hussain, Kayhan Zrar Ghafoor. GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction. Journal of Engineering Research (2025).
3. Tariq, Zhu Wei, Kamran Shaukat*, **Ayesha Aslam**, Adil Hussain, Md. Mahboob Alam**. A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning. Scientific Reports (2026).
4. Adil Hussain*, Vishwanath Eswarakrishnan**, **Ayesha Aslam**, Sajib Tripura. Charging stations demand forecasting using LSTM based hybrid transformer model. Scientific Reports (2025).
5. **Ayesha Aslam**, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, Adil Hussain. Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting. IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS), Bangladesh (2025).
6. **Ayesha Aslam**, Adil Hussain, Noman Iqbal, Muhammad Yasir, Sajib Tripura. Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen (2025).
7. **Ayesha Aslam**, Sajib Tripura, Muhammad Rabiu Abdullahi, Vineet Dhanawat, Adil Hussain. Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer (ViT) Model. 3rd International Conference on Data Science and Emerging Technologies (DaSet), Springer, Malaysia (2024).
8. **Ayesha Aslam**, Adil Hussain, Kashif Naseer Qureshi. Big Data Analytics for AI in Renewable Energy Networks (Chapter 9). Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks, 1st Edition, Elsevier (2025).

SKILLS & EXPERTISE

- **Teaching:** Computer science instruction, lesson and curriculum planning, classroom management, student assessment and feedback, mentoring students (B.Ed-trained; 3+ years of school teaching).
- **Teaching Interests:** Computer Science, Programming, Artificial Intelligence and Machine Learning, Intelligent Transportation Systems.
- **Programming:** Java, C#, C++, C
- **Development:** Android Studio, Visual Studio, NetBeans, MS SQL, MySQL
- **Systems:** Windows, Ubuntu, Kali Linux
- **Research Tools:** EndNote, TeXMaker, MiKTeX, JabRef, Gnuplot, Inkscape
- **Office:** Word, PowerPoint, Excel, OneNote
- **Strengths:** Teaching, research, Android development, HR management
- **Interests:** Sketching and oil painting

FEATURED PROJECTS

- **School Management System (Dar-e-Arqam School, Faisalabad):** built an all-in-one platform for multiple campuses covering admissions, results, fee vouchers and collection, salaries, attendance, and daybook.
- **ADMS, Advocate Diary Management System (Pansoota Law Associates):** delivered a custom MVC case-diary system with hearing schedules, court and client management, payment tracking and SMS alerts, plus a CMS-based public website.
- **SmrtWorx:** designed the app and delivered its functional requirements, including a multi-country currency converter and a money-transfer order processor.
- **Job Portal:** developed a portal for employees, employers, and administrators with freemium and premium accounts, direct hiring, and profile promotion.
- **Assignment Helps:** built a student-support app connecting learners with Australian experts.
- **I Secure:** developed a Java Android application as my final year project.

Links to live projects available on request.

REFERENCES

Available upon request.

APPENDIX: COMPLETE PUBLICATION LIST

List of all journal articles, conference papers, and book chapters

* Corresponding author; ** co-corresponding author.

Journal Articles

2026

1. **Ayesha Aslam**, Xiaojun Yang. Spatial-aware self-attention feature fusion DQN for path planning of USV in static and dynamic environments. Applied Soft Computing (2026), Article 115818, Elsevier.
2. Tariq, Zhu Wei, Kamran Shaukat*, **Ayesha Aslam**, Adil Hussain, Md. Mahboob Alam**. A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning. Scientific Reports.

2025

3. Adil Hussain*, Vishwanath Eswarakrishnan, **Ayesha Aslam**, Sajib Tripura. Charging stations demand forecasting using LSTM based hybrid transformer model. Scientific Reports (2025).
4. **Ayesha Aslam**, Yang Xiaojun*, Kashif Naseer Qureshi**, Adil Hussain, Kayhan Zrar Ghafoor. GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction. Journal of Engineering Research.
5. **Ayesha Aslam**, Z. D. Anuarovna. Real-Time Workload Prediction and Resource Optimization for Parallel Heterogeneous High-Performance Computing Systems Architectures. Ubiquitous Technology Journal, 1(1), 40-47.

2024

6. Adil Hussain*, **Ayesha Aslam**, Sajib Tripura, Vineet Dhanawat, Varun Shinde. Weather Forecasting Using Machine Learning Techniques: Rainfall and Temperature Analysis. Journal of Advances in Information Technology.
7. Adil Hussain*, **Ayesha Aslam**. Hate Speech against Women and Immigrants: A Comparative Analysis of Machine Learning and Text Embedding Techniques. Journal of Applied Research and Technology.
8. Adil Hussain*, **Ayesha Aslam**. Ensemble-based Approach using Inception V2, VGG-16, and Xception Convolutional Neural Networks for Surface Cracks Detection. Journal of Applied Research and Technology.
9. Adil Hussain, Amna Khatoon*, **Ayesha Aslam**, Tariq, Muhammad Asif Khosa. A Comparative Performance Analysis of Machine Learning Models for Intrusion Detection Classification. Journal of Cyber Security.
10. **Ayesha Aslam**, Adil Hussain*. A Performance Analysis of Machine Learning Techniques for Credit Card Fraud Detection. Journal of Artificial Intelligence.
11. Adil Hussain*, **Ayesha Aslam**. Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models. Journal of Artificial Intelligence.

2023

12. Adil Hussain, Asad Ullah*, **Ayesha Aslam**, Amna Khatoon. A Modified Siamese Network for Facial Assimilation. WSEAS Transactions on Signal Processing.

Conference Papers

2026

1. Adil Hussain*, **Ayesha Aslam**. A Reinforcement Learning-Based Adaptive Dynamic Pricing for EV Charging Stations. World Transport Congress (WTC), Tianjin, China.
2. Sajib Tripura, Noman Iqbal, Zahirul Islam Kajal, Adil Hussain, **Ayesha Aslam**, Mostak Ahamed. Traffic Speed Prediction using LSTM and Spatial GCN-based Feature Fusion. IEEE 3rd International Conference on Computing, Applications and Systems, Bangladesh (2026) - ACCEPTED.

2025

3. Sajib Tripura, Adil Hussain, Mostak Ahamed, Dhonita Tripura, **Ayesha Aslam**. A Dual Attention Mechanism-Based Enhanced Residual Network for Skin Cancer Detection. IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS), Bangladesh.
4. **Ayesha Aslam**, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, Adil Hussain. Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting. IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS), Bangladesh.
5. Adil Hussain, Muhammad Yasir, Noman Iqbal, Sajib Tripura, **Ayesha Aslam**. Heart Disease Classification Using Machine Learning Techniques: An Impact Analysis of GridSearchCV-based Optimization. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.
6. **Ayesha Aslam**, Adil Hussain, Noman Iqbal, Muhammad Yasir, Sajib Tripura. Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.

2024

7. Adil Hussain*, **Ayesha Aslam**. A Cost-Aware Optimal Charging Time Algorithm for EVs in a Parking Lot. World Transport Congress (WTC), Qingdao, China.
8. Adil Hussain, Vineet Dhanawat, **Ayesha Aslam**, Noman Iqbal, Sajib Tripura. Credit Card Fraud Detection Using Machine Learning Techniques: Dealing with Imbalanced Data Using Over-Sampling and Under-Sampling Methods. IEEE Beyond Technology Summit on Informatics International Conference (BTS-I2C), Indonesia.
9. Adil Hussain, Kashif Naseer Qureshi, Raja Waseem Anwar, **Ayesha Aslam**. A Novel SCD11 CNN Model Performance Evaluation with Inception V3, VGG16 and ResNet50 Using Surface Crack Dataset. IEEE 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS), Oman.
10. Adil Hussain, Kashif Naseer Qureshi, Faizan Zaman, **Ayesha Aslam**. Minor Surface Cracks Detection using SCD11 Convolutional Neural Network. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.
11. Adil Hussain, Kashif Naseer Qureshi, **Ayesha Aslam**, Tariq, Muhammad Rabiu Abdullahi. TSC18 Convolutional Neural Network for Traffic Sign Classification. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.
12. Adil Hussain, Vineet Dhanawat, **Ayesha Aslam**, Faizan Zaman. Electricity Load Forecasting Using Attention-Based Hybrid Deep Learning Model. IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET), Malaysia.
13. **Ayesha Aslam**, Sajib Tripura, Muhammad Rabiu Abdullahi, Vineet Dhanawat, Adil Hussain. Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer (ViT) Model. 3rd International Conference on Data Science and Emerging Technologies (DaSet), Springer, Malaysia.

14. Faizan Zaman, Zhigang Xu, Adil Hussain, **Ayesha Aslam**, Muhammad Rabiu Abdullahi. Concrete and Pavement Cracks Detection Leveraging Vision Transformer (ViT) Model. IEEE International Conference on Electrical and Information Technology (IEIT), Indonesia.

2023

15. Adil Hussain, **Ayesha Aslam**, Tariq. Traffic Sign Classification Using a Novel TSC18 Convolutional Neural Network. 9th International Conference of Sabaragamuwa University of Sri Lanka (Abstract), Sri Lanka.

Book Chapters

2026

1. Adil Hussain, **Ayesha Aslam**, Mahad Ahmad. Artificial Intelligence in Agriculture: Applications, Challenges, and Future Directions (Chapter 3). AI-Driven Farming Systems With Minimal Human Intervention, 1st Edition, IGI Global, eISBN13: 9798337392974.

2025

2. Adil Hussain, **Ayesha Aslam**, Rundong Wang, Kashif Naseer Qureshi. Advanced Learning Techniques for AI in Renewable Energy (Chapter 7). Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks, 1st Edition, Elsevier, eBook ISBN: 9780443406195.
3. **Ayesha Aslam**, Adil Hussain, Kashif Naseer Qureshi. Big Data Analytics for AI in Renewable Energy Networks (Chapter 9). Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks, 1st Edition, Elsevier, eBook ISBN: 9780443406195.
4. Adil Hussain, Kashif Naseer Qureshi, **Ayesha Aslam**. Novel Machine and Deep Learning and Training Techniques for AI-based Healthcare Systems (Chapter 7). Artificial Intelligence-based Smart Healthcare Systems: New Standards, Technologies and Communication Systems, 1st Edition, Elsevier, Paperback ISBN: 9780443264764.
5. **Ayesha Aslam**, Kashif Naseer Qureshi, Adil Hussain. Advanced AI-Based Healthcare Systems Applications and Services (Chapter 4). Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns, 1st Edition, Taylor & Francis - CRC Press, eBook ISBN: 9781032667911.